Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

FinSight – GenAI Financial Analyst Agent

The domain of the Project:

Generative AI
Automation

Team Mentors (and their designation):

Prof. Radha Kumari (Instructor)
Mr. Prujith Radhakrishanan(Trainer)

Team Member:

Mr. Shubham Yadav

Period of the project:

July 2025 to March 2026

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Declaration

The project titled "FinSight – GenAI Financial Analyst Agent" has been mentored by Prujith Radhakrishanan, organised by SURE Trust, from July 2025 to January 2026, for the benefit of the educated unemployed rural youth for gaining hands-on experience in working on industry relevant projects that would take them closer to the prospective employer. I declare that to the best of my knowledge the members of the team mentioned below, have worked on it successfully and enhanced their practical knowledge in the domain.

Team Members:

Mr. Shubham Yadav

Student, SURE Trust

Prujith Radhakrishnan
Mentor— SURE Trust

Prof. Radhakumari
Executive Director & Founder
SURE Trust

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Executive Summary

FinSight is an AI-powered financial analysis platform designed to automate the extraction and interpretation of data from corporate annual reports. The system leverages a **Retrieval-Augmented Generation (RAG)** architecture to process unstructured financial documents such as 10-K filings and convert them into actionable insights.

The platform enables users to upload company reports and query them in natural language. It intelligently retrieves relevant sections using vector embeddings and generates accurate, context-aware responses supported by **page-level citations**, ensuring transparency and trust.

Beyond basic question answering, FinSight integrates a **financial intelligence layer** that extracts key metrics (revenue, profit, debt), computes financial ratios (P/E, Debt-to-Equity, ROE), and generates structured insights on company performance. It also includes a **comparative analysis feature**, allowing users to evaluate multiple companies side-by-side, and a **rule-based recommendation engine** that provides Buy/Hold/Sell suggestions with explainable reasoning.

By automating manual analysis of lengthy financial reports, FinSight reduces preliminary due diligence time by approximately **60–70%**, making it highly relevant for investment analysts, consultants, and retail investors.

The system is built using **Python, FAISS (vector database), OpenAI-based LLMs, and Streamlit**, with a modular architecture that ensures scalability and extensibility.

Introduction

Background & Context

Financial analysis is a critical function in investment banking, equity research, and corporate finance. Analysts typically spend a significant portion of their time extracting relevant information from large, unstructured documents such as annual reports, which can exceed 200 pages.

With the rise of Generative AI and Large Language Models (LLMs), there is an opportunity to automate this process. However, traditional AI models often suffer from hallucination and lack of contextual grounding.

The introduction of RAG architecture, which combines document retrieval with generative models, provides a reliable solution by ensuring that responses are grounded in actual source data. FinSight is built in this context to bridge the gap between financial domain knowledge and modern AI capabilities, enabling intelligent and trustworthy financial analysis.

Problem Statement / Goals - Financial professionals and investors face challenges in efficiently analyzing large volumes of unstructured financial data. Manual extraction of key metrics, comparison between companies, and interpretation of financial performance require significant time and effort, leading to inefficiencies and delayed decision-making.

- To automate the extraction of key financial metrics from annual reports
- To implement a RAG-based system for accurate and context-aware responses
- To compute financial ratios and generate meaningful insights
- To provide comparative analysis between companies
- To develop a recommendation system (Buy/Hold/Sell) with explainable reasoning

Scope & Limitations

Scope:

- Processing and analyzing **annual reports (PDF format)**
- Extracting financial data such as revenue, profit, debt, and equity
- Performing ratio analysis and generating insights
- Enabling question-answering over financial documents
- Supporting comparison between multiple companies
- Providing decision-support.

Limitations:

- Accuracy depends on the quality and structure of input PDFs
- Financial extraction relies on pattern matching and may not capture all variations

- The recommendation engine is rule-based and does not incorporate advanced valuation models
- LLM responses may still carry minor risks of misinterpretation.
- Performance may vary with very large datasets or highly complex financial statements.
- Does not currently integrate real-time market data or external financial APIs.

Innovation Component -

- Integration of RAG in Financial Analysis
- Citation-Based Answering System
- End-to-End Financial Intelligence Pipeline
- Comparative Analysis Feature
- Explainable Recommendation Engine
- Modular and Scalable Architecture

Project Objectives

1. To develop an AI-based system that can **analyze unstructured financial reports** such as annual reports and extract meaningful information.
2. To implement a **Retrieval-Augmented Generation (RAG) architecture** for accurate, context-aware querying of financial data.
3. To automatically **extract key financial metrics and compute important ratios** for performance evaluation.
4. To enable **comparative analysis between companies** and generate actionable financial insights.
5. To build a **decision-support system** that provides Buy/Hold/Sell recommendations with explainable reasoning and citation-backed outputs.

Expected Outcomes & Deliverables

1. Development of a **fully functional AI-based application** capable of processing financial reports and answering user queries through a user-friendly interface.

2. Implementation of a system that can **extract financial data, compute ratios, and generate insights** for effective performance analysis.

3. A **comparative analysis and recommendation module** that evaluates companies and provides Buy/Hold/Sell decisions with reasoning.

4. Delivery of a **complete, well-structured codebase** with modular architecture, along with supporting test cases for validation.

5. Submission of **comprehensive documentation, project report, and demo presentation** showcasing system design, workflow, and real-world impact.

Methodology and Results

Technologies Used

- Python
- AI / Machine Learning, RAG (Retrieval-Augmented Generation, Vector Database, Streamlit (UI).

Backend Framework

- FastAPI ,PyMuPDF (fitz) ,NumPy ,Pytest ,Git & GitHub,.env (Environment Variables)

Tools/Software

- VS Code, Git/GitHub, API and Testing tools,UI and Deployment tools.

Finsight-- Approach

◈ 1. Problem Understanding & Requirement Analysis

- Identified the challenge of manual financial analysis from lengthy annual reports.

◈ 2. Data Collection & Preprocessing

◈ 3. Embedding & Vector Storage

◈ 4. Retrieval-Augmented Generation (RAG) Implementation

◈ 5. Financial Analysis Engine

◈ 6. Recommendation System

◈ 7. Citation & Explainability Layer

◈ 8. System Integration & UI Development

◈ 9. Testing & Validation

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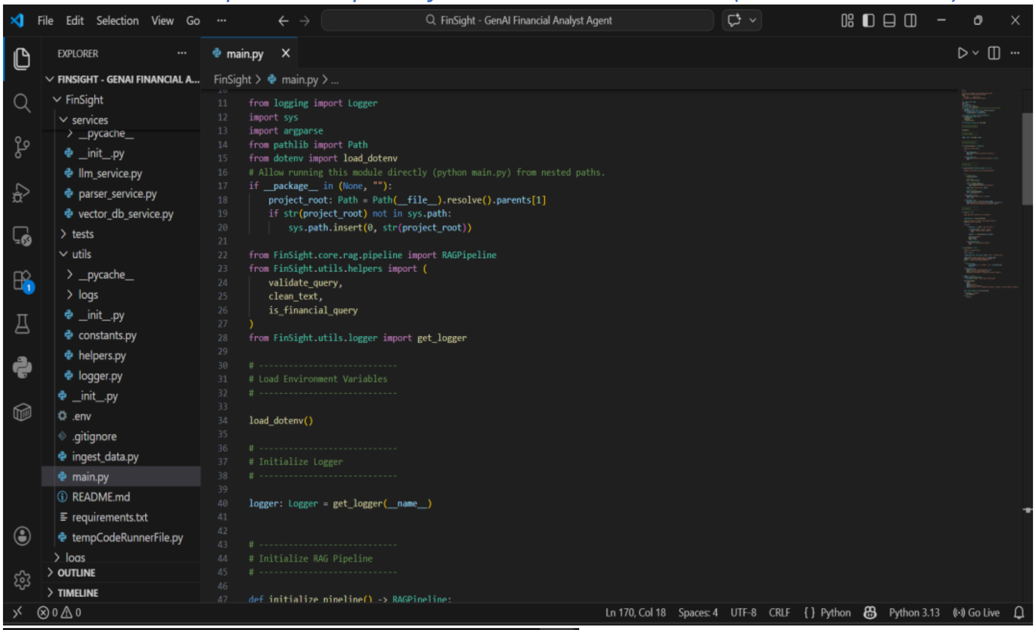
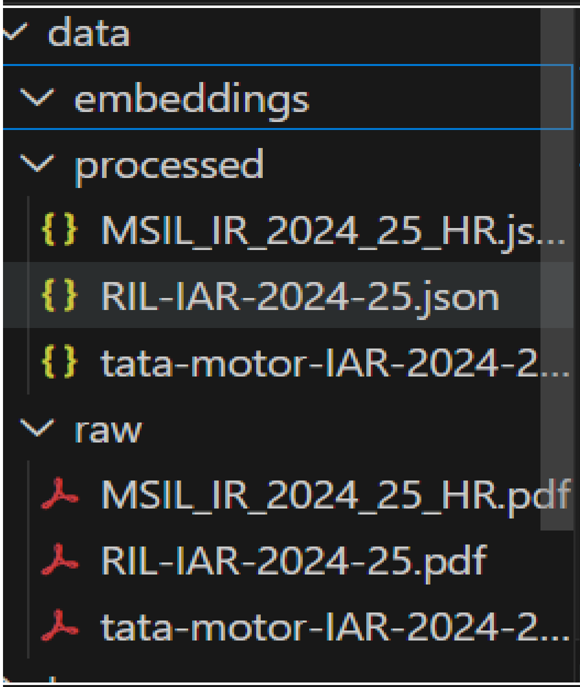
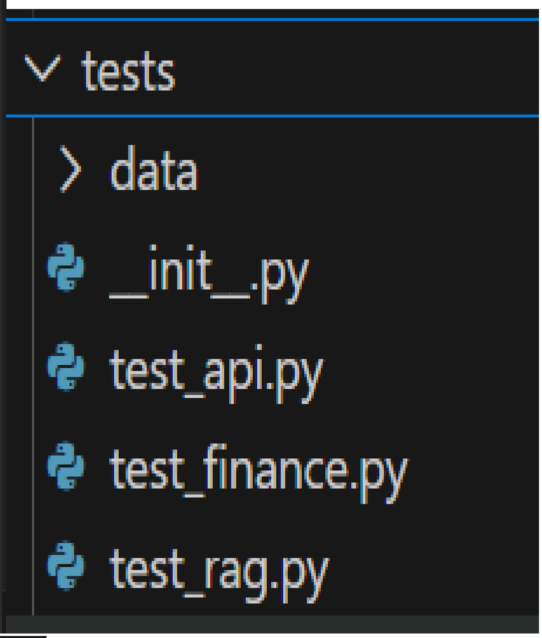
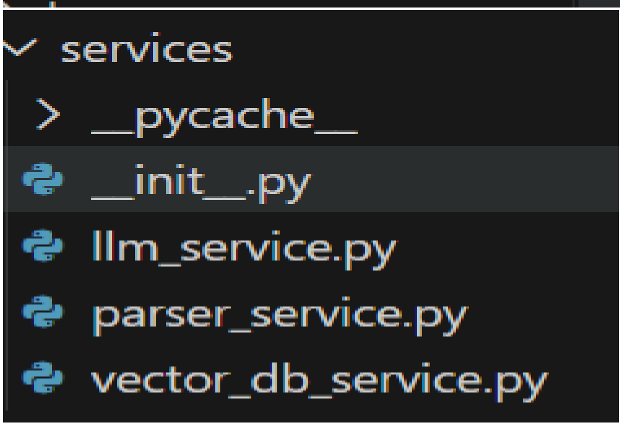
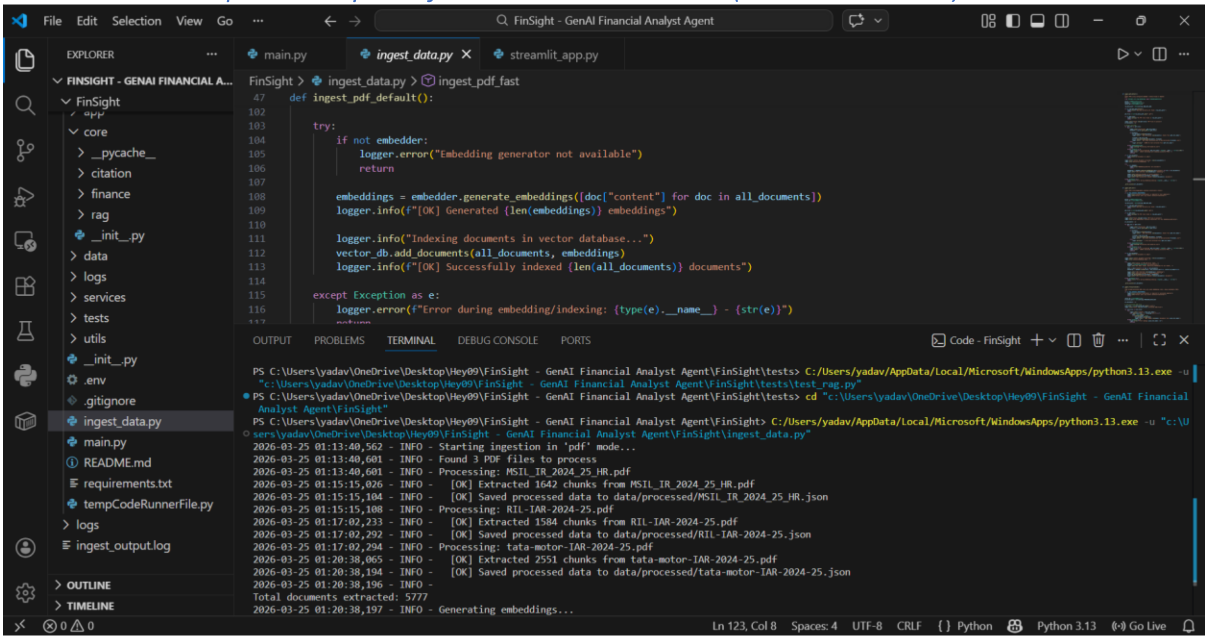
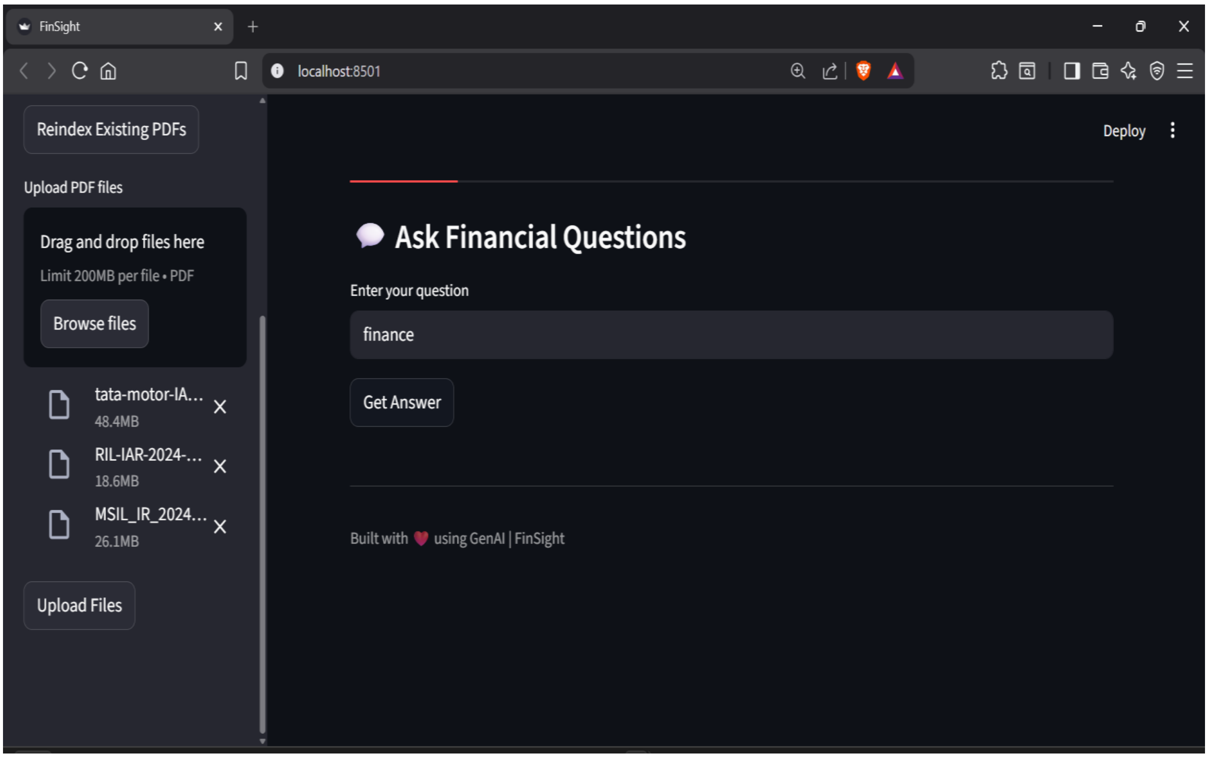









File Edit Selection View Go
FinSight - GenAI Financial Analyst Agent
EXPLORER
FINSIGHT - GENAI FINANCIAL A...
FinSight
core
__pycache__
citation
finance
rag
__init__.py
data
logs
services
tests
utils
__init__.py
.env
.gitignore
ingest_data.py
main.py
README.md
requirements.txt
tempCodeRunnerFile.py
logs
ingest_output.log
OUTLINE
TIMELINE
main.py
ingest_data.py
streamlit_app.py
def ingest_pdf_default():
try:
if not embedder:
logger.error("Embedding generator not available")
return
embeddings = embedder.generate_embeddings([doc["content"] for doc in all_documents])
logger.info(f"[OK] Generated {len(embeddings)} embeddings")
logger.info("Indexing documents in vector database...")
vector_db.add_documents(all_documents, embeddings)
logger.info(f"[OK] Successfully indexed {len(all_documents)} documents")
except Exception as e:
logger.error(f"Error during embedding/indexing: {type(e).__name__} - {str(e)}")
OUTPUT PROBLEMS TERMINAL DEBUG CONSOLE PORTS
2026-03-25 01:13:40,562 - INFO - Starting ingestion in 'pdf' mode...
2026-03-25 01:13:40,601 - INFO - Found 3 PDF files to process
2026-03-25 01:13:40,601 - INFO - Processing: MSIL_IR_2024_25_HR.pdf
2026-03-25 01:15:15,026 - INFO - [OK] Extracted 1642 chunks from MSIL_IR_2024_25_HR.pdf
2026-03-25 01:15:15,104 - INFO - [OK] Saved processed data to data/processed/MSIL_IR_2024_25_HR.json
2026-03-25 01:15:15,108 - INFO - Processing: RIL-IAR-2024-25.pdf
2026-03-25 01:17:02,233 - INFO - [OK] Extracted 1584 chunks from RIL-IAR-2024-25.pdf
2026-03-25 01:17:02,292 - INFO - [OK] Saved processed data to data/processed/RIL-IAR-2024-25.json
2026-03-25 01:17:02,294 - INFO - Processing: tata-motor-IAR-2024-25.pdf
2026-03-25 01:20:38,065 - INFO - [OK] Extracted 2551 chunks from tata-motor-IAR-2024-25.pdf
2026-03-25 01:20:38,194 - INFO - [OK] Saved processed data to data/processed/tata-motor-IAR-2024-25.json
2026-03-25 01:20:38,196 - INFO -
Total documents extracted: 5777
2026-03-25 01:20:38,197 - INFO - Generating embeddings...
Ln 123, Col 8 Spaces: 4 UTF-8 CRLF Python Python 3.13 Go Live


FinSight
localhost:8501
Reindex Existing PDFs
Upload PDF files
Drag and drop files here
Limit 200MB per file • PDF
Browse files
tata-motor-IA...
48.4MB
RIL-IAR-2024-...
18.6MB
MSIL_IR_2024...
26.1MB
Upload Files
Deploy
Ask Financial Questions
Enter your question
finance
Get Answer
Built with using GenAI | FinSight

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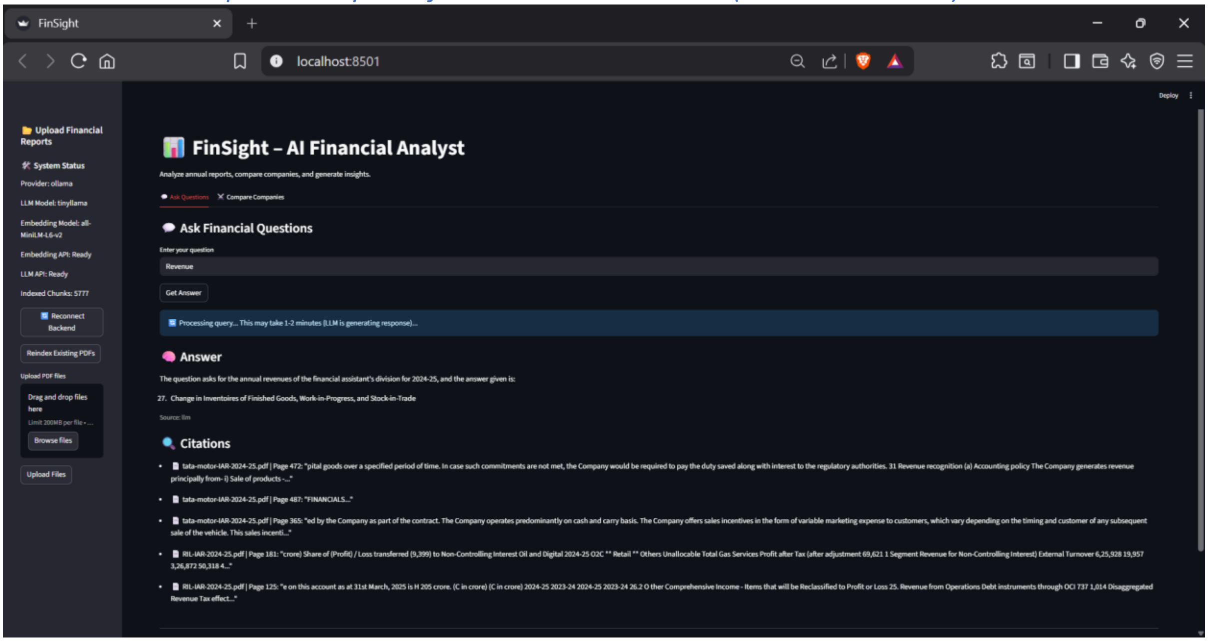

GitHub - [Link]()

if any problem occurrs refer to - https://github.com/sure-trust/SHUBHAM-YADAV-g2-genai

Learning and Reflection

Gained practical understanding of **how AI can be applied to real-world financial problems**, beyond theoretical concepts.

Learned the implementation of **Retrieval-Augmented Generation (RAG)** and how it improves accuracy over traditional AI models.

Developed skills in **processing unstructured data**, especially extracting meaningful information from large PDF documents.

Improved knowledge of **financial concepts** such as revenue, profit, debt, and key financial ratios.

Understood the importance of **data preprocessing and chunking** in improving model performance.

Learned how to integrate multiple components like **embeddings, vector databases, and LLMs** into a single system.

Gained experience in building a **modular and scalable architecture**, making the system easy to extend.

Understood the importance of **explainability and trust** in AI systems through citation-based responses.

Improved debugging, testing, and problem-solving skills while working on **end-to-end system integration**.

Realized the limitations of AI systems and the need for **continuous improvement and validation** in real-world applications.

Conclusion and Future Scope

Recap of Objectives & Achievements

FinSight successfully demonstrates how **Generative AI can be applied to financial analysis** to automate the processing of unstructured data such as annual reports. The system integrates a **Retrieval-Augmented Generation (RAG)** architecture with financial intelligence to provide accurate, context-aware, and citation-backed insights.

By enabling natural language interaction, extracting key financial metrics, performing ratio analysis, and generating recommendations, the project significantly reduces the time and effort required for manual financial analysis.

Overall, FinSight acts as an effective **decision-support tool** that bridges the gap between complex financial data and user-friendly insights, making financial analysis more efficient, accessible, and reliable.

Future Scope

- Integration of Real-Time Data
- Advanced Financial Modeling
- Improved Data Extraction
- Multi-Document & Multi-Year Analysis
- Enhanced Recommendation Engine
- User Personalization
- Scalability & Deployment
- Visualization Dashboard

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